

The Selected $K3$ Surfaces of the Cooper s_7/s_{10} Families: Identification, the $K3 \times T^2$ Reading, and the Twisted-Weierstrass Route

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Stream 2 — Selection & Geometry (SocrateAI-Scientific-Agora-K3-DarkMatter)

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What is not claimed

This is a mathematics note. It makes no physical claim: no dark-matter identification, no dark-energy statement, no observable. No point of either family maps to any measurement, the flux/tadpole route stays blocked until a threefold base is specified, and the physical reading of “ $K3 \times T^2$ ” stays **Tier C — blocked physics**. Nothing here ranks the s_7 family against the s_{10} family. No Kodaira fibre type is attached to any point of either family; we report discriminant-root orders and root lattices only.

Abstract

The 2026-09-29 note adopted, by an explicit ruling, a within-family selector on the $\rho = 20$ points of the Cooper s_7 and s_{10} Picard–Fuchs families: the point of minimal $|\text{disc } T|$. This note records what happened next. First, the selected points are identified as named surfaces. Through the Shioda–Inose correspondence between singular $K3$ surfaces and even positive-definite binary forms, and Takatsu’s tabulation, the s_7 pick ($T = A_2$, $\det T = 3$) is the unique singular $K3$ surface X_3 and the s_{10} pick ($\det T = 4$) is the unique X_4 . The pair is the one called “Vinberg’s two most algebraic $K3$ surfaces” in the Schütt–Shioda survey. Second, we state the one reading of “ $K3 \times T^2$ ” under which a selection is meaningful, the Shioda–Inose pairing with a pair of n -isogenous elliptic curves, and report the unscored consistency check between the level read from the lattice and the level read from the modular side. Third, we record the twisted-Weierstrass route. A degree question that had conflated two quantities is resolved, a two-divisor degree screen on candidate threefold bases is reported, and the Mordell–Weil section required of the M_7 -polarized member is computed. It has height 14 and $\bar{P} \cdot \bar{O} = 5$ generically and at $z = 1/27$, and $\bar{P} \cdot \bar{O} = 0$ at $z = -1$. The selected point $z = \infty$ has Mordell–Weil rank 0 and carries no such section. At $z = 1/27$ an explicit endomorphism of square -7 is written over $\mathbb{Q}$. Every table is generated from a certificate.

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1 Scope

Ledger rules of this programme bound everything below. The $\rho = 20$ cut and the discriminant-floor selector are naming conventions inside one family, adopted by explicit rulings; they do not compare the two families. The agreement between the binary-form side and the modular side of any statement here is forced by the Shioda–Inose structure (an elliptic point of the modular curve is a CM point) and is never offered as corroboration. The s_{10} lattice certificate is LIVE since 2026-09-29; that changes no value below and ranks nothing. The next step of the twisted-Weierstrass route that would involve a threefold (work package TW3) has not been opened, because no ruling specifies twist data over a threefold base.

2 The selection, in one table

Table 1 repeats the six $\rho = 20$ points of the 2026-09-29 note, regenerated from the current certificates. The lattice arithmetic of every row is kernel-checked in Lean 4 (Tier A); the identification of the orthogonal complement with the family’s transcendental lattice T is Tier B — derived, not measured (Dolgachev §7, Doran Thm. 5.13, Tier L — literature). The selector picks, in each family, the row of minimal $|D|$: $z = \infty$ in both.

Family	z	v	v^2	D	T (reduced)	Tier / sources
cooper_s7	-1	$(2, -4, 1)$	-2	-7	$[2, 1; 1, 4]$	Tier A, 2 source(s)
cooper_s7	$1/27$	$(1, -1, 0)$	-2	-28	$[2, 0; 0, 14]$	Tier A, 2 source(s)
cooper_s7	∞	$(14, -14, -5)$	-42	-3	$[2, 1; 1, 2]$	Tier A, 2 source(s)
cooper_s10	$-1/4$	$(2, -6, 1)$	-4	-20	$[4, 2; 2, 6]$	Tier A, 1 source(s)
cooper_s10	$1/16$	$(1, -1, 0)$	-2	-40	$[2, 0; 0, 20]$	Tier A, 1 source(s)
cooper_s10	∞	$(10, -10, -3)$	-20	-4	$[2, 0; 0, 2]$	Tier A, 2 source(s)

Table 1: The six $\rho = 20$ points. Compared with the 2026-09-29 build, the s_{10} rows no longer carry an “(ADVISORY)” tag; see section 7, item 1.

3 Which surfaces were selected

A singular $K3$ surface is one of Picard number 20. Shioda and Inose, in the form we read in Shioda’s survey and in Kumar–Kuwata (Thm. 8.1), give a bijection between singular $K3$ surfaces up to isomorphism and even positive-definite binary lattices up to $\mathrm{SL}_2(\mathbb{Z})$, the surface going to its transcendental lattice. A selected point is therefore named by its T . Takatsu’s paper on the singular $K3$ surfaces of discriminant 3, 4 and 7 states that each of these discriminants carries exactly one surface, called X_3 , X_4 , X_7 , and tabulates their models. Our checker parses that

table and the uniqueness sentence from the pinned text and, independently, enumerates the reduced forms of each determinant.

Family	z	T (Gram)	$\det T$	classes	Surface
cooper_s7	∞^*	$[2, 1; 1, 2]$	3	1	X_3
cooper_s7	-1	$[2, 1; 1, 4]$	7	1	X_7
cooper_s7	$1/27$	$[2, 0; 0, 14]$	28	2	not named by $\det T$ alone
cooper_s10	∞^*	$[2, 0; 0, 2]$	4	1	X_4

Table 2: Identification of the s_7 points and the s_{10} pick. *: the point picked by the adopted selector. “classes”: the number of reduced even positive-definite forms of that determinant, computed. At $z = 1/27$ two classes share $\det T = 28$, so the determinant alone names no surface there.

Headline result

The s_7 pick is the singular $K3$ surface X_3 ($T = A_2$, $\det T = 3$); the s_{10} pick is X_4 ($T = \langle 2 \rangle \oplus \langle 2 \rangle$, $\det T = 4$). These are the two smallest determinants any even positive-definite binary lattice attains. The s_{10} family cannot reach $\det 3$, because A_2 does not occur in it (a lattice fact recorded in an earlier certificate, not a preference). Tier B — derived, not measured for the identification of T ; Tier L — literature for the bijection and Takatsu’s table.

Remark 3.1. The name “Vinberg’s two most algebraic $K3$ surfaces” for the surfaces of discriminant 3 and 4 is taken from the Schütt–Shioda survey (Tier L — literature). Vinberg’s 1983 paper itself was not fetched or read. The explicit M_7 -polarized model of the 2026-09-29 note gives $E_\omega \times E_\omega$ at $z = \infty$, which matches Takatsu’s construction of X_3 from $E_3 \times E_3$; this match is forced by the Shioda–Inose structure and is reported as consistency, not as independent evidence.

4 What “ $K3 \times T^2$ ” can mean here

Two readings were put to the decision authority on 2026-09-16. Under the literal product reading, the T^2 factor is a fixed torus unrelated to the $K3$, and every $K3$ surface serves equally well, so no selection follows from it. Under the Shioda–Inose reading, “ T^2 ” is the elliptic side of the pairing the programme has certified: the order-three operator is the symmetric square of an order-two operator, and a member of Picard number 19 is paired with a product $E \times E'$ of two elliptic curves related by a cyclic isogeny of degree n , with $T \cong U \oplus \langle 2n \rangle$. Selection then becomes a question about n , which can be checked. The 2026-09-16 proposal adopted this reading and the check below presupposes it, but the choice between the two readings has *not* been ruled by the decision authority (see section 7, item 6). The isogeny statement is standard; its primary source (Morrison, 1984) was not fetched for this note.

The check adopted for this reading is unscored: it compares the level n read from the lattice certificate (Gram determinant and a serialized U -splitting witness, both recomputed) with the level read from the modular side (an eta-quotient Hauptmodul for $\Gamma_0(n)^+$ or $\Gamma_0(n)^*$, with Newman’s conditions and an Atkin–Lehner check). Table 3 gives the result. Its power to fail rests on real negative cases: four order-three entries of the same register fail the modular leg.

Family	n from lattice	n from modular side	Verdict
cooper_s7	7	7	T3_AGREE(n=7)
cooper_s10	10	10	T3_AGREE(n=10)

Table 3: Lattice level against modular level. An unscored consistency gate, never a ranking.

The physical reading of $K3 \times T^2$ as a compactification is **Tier C — blocked physics** and is not used. A separately submitted document linking $K3 \times T^2$ to Mathieu moonshine was checked claim by claim before this note was written: of twelve integer-sequence identifiers it cited, seven were fabricated or attached to the wrong object. Moonshine plays no part here; in any case the $K3$ elliptic genus is the same for every $K3$ surface, so it cannot separate candidates.

5 The twisted-Weierstrass route

A strict-pullback realization of the s_7 family as a Calabi–Yau threefold over the bases tried was excluded earlier (a structural negative, **Tier B — derived, not measured**). The route kept open is a twisted Weierstrass model whose $K3$ fibres carry the M_7 polarization. Three steps have results.

5.1 TW0: a degree question that conflated two quantities

An earlier plan set a fundamental line-bundle degree $\ell = 2$ “via the symmetric-square theorem”. Recomputing from the operator’s Riemann scheme separates two numbers (table 4). The degree of the Weierstrass model’s fundamental line bundle is $\chi(\mathcal{O}_{K3}) = 2$, from Noether’s formula with $\deg \Delta = 24$ (**Tier A** -level arithmetic, standard). The family-level Hodge bundle over the orbifold curve $X_0(7)^+$ has degree $1/3$, and its square has degree $2/3$ (**Tier B — derived, not measured**). The earlier formula reproduced 1 from a Fuchs-relation identity. The rule now reads $\ell := \chi(\mathcal{O}_{K3}) = 2$.

Quantity	Value
orbifold Euler characteristic of $X_0(7)^+$	$-2/3$
$\deg \omega$ (family Hodge bundle, $\mathbb{Q}$ -degree)	$1/3$
$\deg \omega^2$	$2/3$
$\chi(\mathcal{O}_{K3})$ (elliptic-surface reading of ℓ)	2
$\deg \Delta$ of the Weierstrass model	24

Table 4: The two degrees that the earlier plan had merged, from the s_7 operator’s exponents.

5.2 TW1: a necessary screen on threefold bases

The fibres of the generic member contain two E_8 root configurations. A twisted model over a threefold base B_3 needs two divisors carrying the corresponding Tate vanishing orders ($v(f) \geq 4$, $v(g) \geq 5$, $v(\Delta) = 10$) inside the degree budget set by $-4K$, $-6K$ and $-12K$. Table 5 lists the result. On $\mathbb{P}^3$ the budget alone is met, but any two effective divisors meet in a curve, along which the orders reach $(8, 10, 20)$ and the model is not minimal; the screen fails. On $\mathbb{P}^1 \times \mathbb{P}^2$ and on the projective bundles $\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(n))$ over $\mathbb{P}^2$, two disjoint divisors exist and the screen passes. Passing is a necessary condition only: nothing here constructs a model.

Base B_3	Two-divisor degree screen
$\mathbb{P}^3$	FAIL
$\mathbb{P}^1 \times \mathbb{P}^2$	PASS
$\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(n))$ over $\mathbb{P}^2$	PASS (degree budget all $n \geq 0$; realizability checked $n \leq 18$)

Table 5: Two-divisor degree screen. The certificates were written on 2026-07-29 and their status string still reads “DRAFT”; the screen was made live by ruling D20’ on 2026-10-07 (section 7).

5.3 TW2: the section that the M_7 polarization requires

On a $K3$ surface with an elliptic fibration having two E_8 root configurations, the trivial lattice has rank 18 and discriminant ± 1 . Reaching $|\text{disc NS}| = 2n$ with Picard number 19 then needs one Mordell–Weil section P with height $h(P) = 2n$. The height formula (Schütt–Shioda Thm. 11.5) gives $h(P) = 2\chi + 2\bar{P} \cdot \bar{O} - \sum \text{contr}_v(P)$, so with $\chi = 2$ and no contributions, $\bar{P} \cdot \bar{O} = n - 2$: for s_7 , $h = 14$ and $\bar{P} \cdot \bar{O} = 5$. Kumar–Kuwata give an independent route for $E \not\cong E'$: the section attached to an isogeny of degree d has height $2d$, and $d = 7$ gives the same 14.

At the three $\rho = 20$ points the two curves become isomorphic and the count changes. Exact reduction of the explicit model at each point gives the extra root lattice; the height formula then fixes the section (table 6).

s_7 member	$ \text{disc } T $	ρ	extra roots	MW rank	$h(P)$	$\bar{P} \cdot \bar{O}$
generic	14	19	—	1	14	5
$z = 1/27$	28	20	A_1	1	14	5
$z = -1$	7	20	A_1	1	7/2	0
$z = \infty$	3	20	A_2	0	—	no section

Table 6: The section required by the M_7 polarization: generic member and the three $\rho = 20$ points of s_7 . Root lattices from exact reduction of the explicit model; heights from Schütt–Shioda Thm. 11.5 (Tier B — derived, not measured).

Headline result

The section with $\bar{P} \cdot \bar{O} = 5$ exists on the generic member and at $z = 1/27$. At $z = -1$ the extra A_1 absorbs part of the height and $\bar{P} \cdot \bar{O} = 0$. At $z = \infty$, the point the selector picked, the extra lattice is A_2 , the Mordell–Weil rank is 0 and there is no section at all. The selected surface X_3 therefore does not carry the twisted-route section; the selection and the route are separate questions.

At $z = 1/27$ the elliptic curve has $j = 16581375$ and complex multiplication by the order of discriminant -28 . Using Vélu’s formulas over $\mathbb{Q}$ we wrote an explicit 7-isogeny φ from this curve to a curve with the same j , and checked $\varphi \circ \varphi$ against multiplication by 7 at 60 rational points, more than the degree bound of 49. The x -coordinates fix $\varphi \circ \varphi$ only up to sign, $\pm[7]$. The sign is -7 because the endomorphism algebra is the imaginary quadratic field of discriminant -28 , which has no element of square $+7$ (Tier B — derived, not measured , through the CM certificate). The remaining step, carrying the graph of φ through the Kumar–Kuwata construction down to the $K3$ surface and reading $\bar{P} \cdot \bar{O} = 5$ on the pinned model, is specified but not done.

6 What this note gives the dark-matter stream

Nothing physical. The selected surfaces X_3 and X_4 , the $K3 \times T^2$ reading and the section computation are mathematics about lattices and elliptic surfaces; no CM point and no selected surface maps to any dark-matter observable, and no number here may be used as a prior, a parameter or a fit input. The companion Stream 3 note (repository [DarkMatterK3-Home.github.io](https://github.com/DarkMatterK3-Home), 2026-10-08) reports calibration checks and a literature recount for a mixed fuzzy-dark-matter search window; it uses no result of this note.

7 Corrections recorded in band

Corrections recorded in band

1. The table generator of the 2026-09-29 note attached “(ADVISORY)” to the s_{10} rows by family name. The PDF rebuilt on 2026-10-07 therefore still printed three tags, while its own dated correction said the tables showed none. The generator now reads each row’s `advisory_family` flag (false for every row), a control fails if the tag is keyed on anything else, and that note carries a dated item 7.
2. The same generator typed the discriminant-root orders of its fibration table as literals, against the rule that numbers are computed. The printed values were right; they are now read from the certificate, and a tampered order changes the table (control).
3. On 2026-10-07 a record stated that $\varphi \circ \varphi = [-7]$ was proved. The computation fixed only $\pm[7]$; the sign comes from the CM discriminant. The record carries a dated correction.
4. The author of the discriminant 3, 4, 7 paper was first taken from a search-result snippet, which named other authors; it was corrected from the paper’s own header before any certificate was emitted.
5. The TW1 certificates still read “DRAFT – pending” although ruling D20’ made the screen live. They are left unchanged, as provenance; their status is the one the ledger records.
6. *Dated correction, 2026-10-08, after release v0.3.17.* The first release of this note called the Shioda–Inose reading of $K3 \times T^2$ the one “which this programme uses”. That reading was adopted by a proposal and is presupposed by the unscored T3 check, but the decision authority has not ruled between it and the product reading; the 2026-09-21 “decision 1” concerned the $\rho = 20$ cut, not this choice. Section 4 now says so.
7. *Dated correction, 2026-10-08.* The repository’s regression list contained paper-lint lines (for the 2026-09-29 note since that date, and for this one), but the continuous-integration runner executes its own fixed list and never read them, so no paper had been linted in CI. The table check and the paper lint are now in the runner itself. The first release of this note was linted locally only.

8 Reproducibility

Every table is generated from `data/certificates/` by `scripts/render_paper_tables.py`, whose `-check` mode fails if any fragment drifts from its certificate; its controls include tampered inputs for every new table. The prose was linted by `scripts/check_paper_tier_language.py`, which applies the programme’s tier-language rules to the whole body and blocks Kodaira fibre-type tokens; it ships a self-test with planted violations. Decision records (D19’–D28’) and the identification brief:

- `briefs/TO_DECISIONS_2026_10_07_STREAM2_DELEGATED.md`
- `briefs/SELECTED_K3_IDENTIFICATION_2026_10_08.md`

Provenance

Generated-by: Claude (Opus 5.5), Stream 2 | Verified-by: `render_paper_tables.py -check` and its controls; each cited checker's own controls | Reviewed-by: N (preprint; not reviewed).

References

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- [4] M. Schütt and T. Shioda, “Elliptic surfaces,” *arXiv:0907.0298*.
- [5] I. Dolgachev, “Mirror symmetry for lattice polarized $K3$ surfaces,” *arXiv:alg-geom/9502005*.
- [6] C. F. Doran, “Picard–Fuchs uniformization: modularity of the mirror map and mirror-moonshine,” *arXiv:math/9812162*.
- [7] A. Clinger, C. F. Doran, J. Lewis, U. Whitcher, “Normal forms, $K3$ surface moduli, and modular parametrizations,” *arXiv:0712.1880*.
- [8] X. Callens and the SocrateAI Scientific Agora Collaboration, “Certified lattice data and a within-family selector for the Cooper s_7/s_{10} $K3$ families,” Stream 2 note, 2026-09-29 ([papers/stream2_selection_geometry_2026_09_29.pdf](#)).